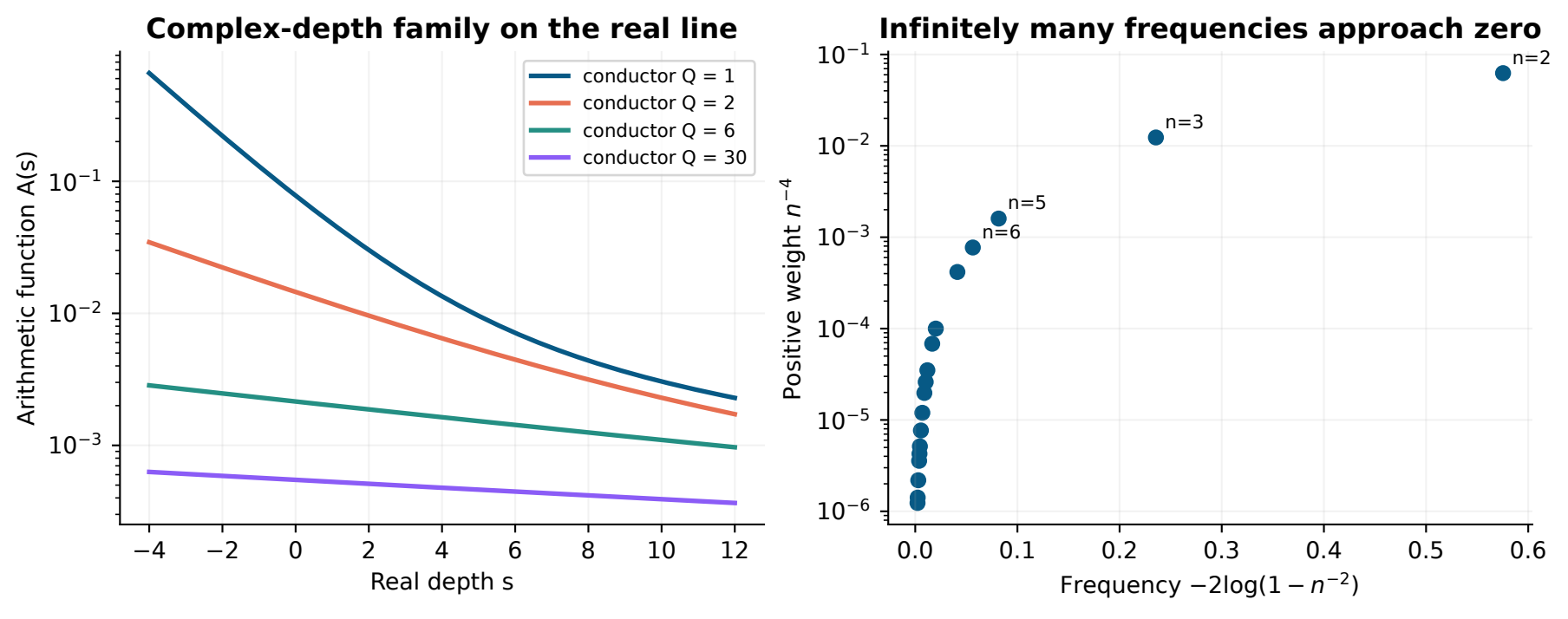
Complex-depth family on the real line
conductor Q = 1
conductor Q = 2
conductor Q = 6
conductor Q = 30
Arithmetic function A(s)
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
6
8
10
12
Real depth s
Infinitely many frequencies approach zero
Positive weight n^{-4}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
n=2
n=3
n=5
n=6
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Frequency $-2\log(1-n^{-2})$